Communication Intelligence

How the system works, how it gets smarter, and how it stays accurate.

FINISHES SOLUTIONS · SYSTEM DIAGRAM

Source Microsoft 365 email, every mailbox in scope
Store PostgreSQL and pgvector in the Finishes Azure tenant
Ask from Claude, Microsoft Copilot or ChatGPT, signed in as yourself

01 How it works

Every email follows the same seven steps. Nothing is searchable until it has passed steps 3 and 4.

STEP 1

Mailboxes

- **Every company mailbox** is in scope, user and shared; named exceptions are kept out.
- Access is **read-only** and granted mailbox by mailbox through Exchange's own permission boundary.
- The system cannot send, move or change mail.

176 MAILBOXES TODAY

STEP 2

Capture

- Each email is fetched with its **subject, body, participants and attachments**, using IDs that survive moves.
- Attachments are opened and their **text extracted**; original files are kept.
- Obvious bulk and automated mail is recorded, not processed.

~19,000 INDEXED EMAILS A MONTH

STEP 3

Protect

- **Masked before storage:** government IDs, bank, routing and card numbers, passwords, keys and codes.
- **Health information** is blocked entirely and never indexed.
- Text is screened for **hidden instructions**; email content is always treated as untrusted.

IN-TENANT SERVICES, NO PROMPT RULES

STEP 4

Understand

- The AI answers **21 fixed questions** per email: action required, approval, invoice, project risk, urgency, category.
- It reads out **properties, entities, people and facts** as written, never inferred.
- A second check asks **whose** action item it is.

LABELS ARE PROPOSALS, NOT APPROVALS

STEP 5

Store

- Text is cut into passages and turned into **meaning vectors** for search by idea, not just keyword.
- Everything lives in **PostgreSQL in the Finishes Azure tenant**; files in private Blob storage.
- Copies of one email in several mailboxes are linked, never merged.

ONE DATABASE, ONE AUDIT TRAIL

STEP 6

Ask, as yourself

- You ask from **Claude, Copilot or ChatGPT** after a Microsoft 365 sign-in.
- The assistant reaches mail only through **one authorized API**. It never holds a database key.
- Your identity decides what you can see, **on every question**.

SEARCH · THREADS · ACTION ITEMS · RISKS

STEP 7

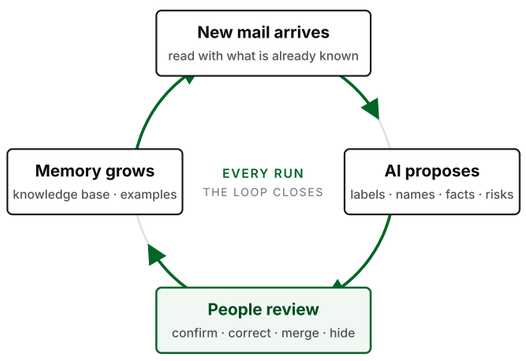
Get the evidence

- Every answer cites **sender, date, subject and mailbox** and links to the real email and its files.
- An empty result means **"not found in mail you may read"**, never a guess.
- Links open a read-only page under the same access rule.

ADMIN CONSOLE FOR REVIEW, COSTS, PEOPLE

02 How it gets smarter

Two loops, both corrected by people, never approved by the AI.



A **A knowledge base learned from the mail itself**
Properties, companies, people, terms, meetings and how they relate are extracted from each email and checked against its text. A name seen in two or more messages is handed back to the AI as context for the next one, so references get resolved the way the company actually writes them. No seeded list, no approval queue.

B **Reviewer answers become examples**
In Label Review, a person marks each AI label right or wrong. Corrected emails are shown to the AI as worked examples on later calls, corrections first, then mail from the same sender domain. Precision and recall per label come from the same answers.

C **Risk picture accumulates**
Each project-risk email gets an impact and likelihood rating with a stated reason. Over time this builds the risk matrix, the trend by category and the property heatmap.

D **Thresholds move only by hand**
The system suggests where a label's yes/no line should sit. A person changes it. Nothing retunes itself silently, so behaviour is explainable at any date.

03 How it stays accurate

Rules that are enforced, and results that are measured, not asserted.

Enforced on every email and every question

Masking is a service, not an instruction
Sensitive numbers are found by deterministic patterns and an in-tenant detection service before the AI ever reads the text. A masking outage holds the email rather than letting it through.

Email text is untrusted input
Instructions found inside a message are ignored by the classifier and by every assistant. Injection attempts are screened and flagged.

Nothing is silently dropped
Every email ends in a recorded state: published, held with a reason code, not indexed, or failed. Operators see counts, backlog and failures per run.

Authorization runs inside the database
Who is asking is checked in SQL before any ranking. Per-reader limits (a person, a domain, a term) only narrow results and apply to every tool and link alike.

Every result is traceable
Sender, date, subject, mailbox and a link to the original. Summaries and labels are shown as AI proposals. Duplicates are collapsed only after authorization.

Every call is audited
Who asked, which tool, which emails came back. Query text is not logged by default. Access changes are written to the audit trail as well as to Entra.

MEASURED SEPARATELY, WITH COUNTS

Capture completeness — PER MAILBOX
Mail found in the mailbox versus mail captured. Misses are counted, not assumed away.

Processing success — PER RUN
Published, held by reason, failed and dead, plus the age of the oldest waiting email.

Label quality — PRECISION · RECALL
Per label and per person, from Label Review answers, always with the sample size. A probability is not an accuracy.

Retrieval quality — RECALL@5 · @10
Against a question set with known answers, including questions that should return nothing.

Authorization negatives — MUST BE ZERO
Wrong audience, missing role, revoked grant, excluded mailbox, paused mode, forged filter: each must return no results.

Accuracy is a measured, published number with a sample size, never a single percentage. Thresholds are uncalibrated until a labelled sample says otherwise, and the console shows that status.

Mail never leaves the Finishes Microsoft and Azure tenants for processing. Assistants receive only authorized results, never credentials, keys or files.